

Research Proposal:

**The impact of AI on the development of critical thinking
and problem-solving skills in university students**

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Introduction

Artificial Intelligence (AI), with its growing impact on contemporary society, has emerged as a popular topic. Number of questions about how AI affects critical thinking and problem-solving abilities, particularly in university students, remain unsolved despite substantial research into the effects of AI on human existence. The impact of AI on the growth of these talents in university students must therefore be fully understood through extensive research.

The primary aim of this research is to investigate how AI influences university students' capacity for critical analysis and problem-solving, while also acknowledging that AI can have positive as well as negative impacts on these abilities. While the objectives of this research are:

- To review the existing literature and provide a more detailed and analytical discussion on the impact of AI on critical thinking and problem-solving skills in university students.
- To explore the benefits of implementing AI and its tools in education, particularly for university students.
- To examine how the integration of AI into teaching practices can help university students enhance their critical thinking and problem-solving abilities.

To achieve this goal, the study will examine how AI may improve academic learning environments through personalized instruction, immediate feedback, and enhanced data analysis. AI has had a big impact on the field of education and has the power to significantly change how students learn and develop critical abilities. It is true that AI is changing the way students learn and acquire essential skills. Now, there are availability of different techniques and resources to learn. This study will look at how AI is being used in educational institutions, with an emphasis on teaching and learning and the positive impacts that it has on education. It will also look at potential drawbacks, such as an increasing dependence on AI, and identify and address them.

Furthermore, the study will also examine the various strategies and techniques used in integrating AI into teaching and learning. This study can offer insights on how AI can be efficiently used to enhance critical thinking and problem-solving skills by studying how AI is integrated.

In this research, following questions will be answered:

1. What are the potential implications of AI on university students' critical thinking and problem-solving abilities?
2. How can AI be integrated into teaching practices to improve the development of critical thinking and problem-solving skills in university student?
3. What are the potential benefits of using AI in education?

In general, this research attempts to provide a holistic picture of the impact of AI on university students' growth in critical thinking and problem-solving skills, considering both the benefits and implications of AI integration in the field of education.

Literature Review

Over the past few years, interest in the impact of artificial intelligence (AI) on education has grown significantly, particularly in relation to developing students' critical thinking and problem-solving skills. The application of AI in education has opened up new possibilities for enhancing cognitive skills in students, such as problem-solving and critical thinking, as well as the learning process itself. As AI technology evolves at a rapid rate, educators are increasingly exploring how they can integrate AI into teaching and learning strategies to enhance students' cognitive abilities. The purpose of this review of the literature is to research existing literatures thoroughly and look into how AI has affected how university students further develop their critical thinking and problem-solving abilities as well as its implications and integrations.

According to the research (Zhu & Ren, 2022), it is critical to investigate how to use AI to successfully help teaching and learning as it continues to grow. This paper examines how artificial intelligence (AI) affects role cognition in Taiyuan City, Shanxi Province, Central China's educational system using a questionnaire survey approach. Preschool education, primary education, secondary education, higher education, adult education, computer network education, enterprise education, and social education are the eight categories that make up the targeted education system in this questionnaire study. A total of 34 questions, with a 5-level score, are included in the questionnaire's design, which are broken down into six role cognition items. In all of the aforementioned educational groups, 368 teachers, 402 students or learners, 118 school administrators, and 124 relatives of students or learners make up the responses. The questionnaire included basic information, additional questions, and open-ended questions. A total of 1012 questionnaires were distributed; 978 of them were recovered, recovering at a rate of 96.64%, and 957 of them were valid, recovering at a rate of 97.85%. The data were analyzed statistically using SPSS 19.0 software. Then, the findings were examined using t-tests and one-way analysis of variance. The self-made and open-ended questions were evaluated using qualitative research analyses.

The results of the study show that role cognition is an important aspect of learning in AI-assisted courses, and that developing role cognition necessitates knowledge of the course's subject matter, instructional methods, and activity kinds. The influence of AI on role cognition in the educational system must thus be carefully examined from the viewpoints of function realization form, resource presentation method, supporting hardware form, teacher-student interaction form, and representation method of works. However, as connectors, teachers are limited in their ability to monitor and modify based on how well they understand their pupils, how many resources they have at their disposal, and how effectively they can manage data. There is no consensus on the language to define AI's role in role recognition, despite the fact that it is flexible and diversified, works in learning and teaching activities in a variety of ways.

However, although AI is becoming more prevalent in the educational system, it is not effective at teaching complicated concepts or emotions. Teachers should therefore be confident in their capacity to serve as instructors. There is still a significant distinction between face-to-face engagement and face-to-face communication. The impact of AI is significantly less than what students can contribute to the development of propositional knowledge. AI can analyze students' emotions, behavior, and learning habits as well as track and record their progress. This knowledge can be used by teachers to give students specific direction and help. It's critical that teachers continue their education and embrace new technology. Additionally, they ought to design a learning environment that supports each student's development of critical thinking abilities and creativity. While AI can assist with some activities, it cannot take the place of human invention and learning.

Similarly, based on the research paper (Alharbi et al., 2022), how e-collaborative learning affects the growth of higher order thinking abilities and critical thinking is investigated thoroughly by relating e-collaborating learning to integrated AI learning. According to the research (Alharbi et al., 2022), 200 female kindergarten students were enrolled in a course called "Research Seminar" in order to study critical and higher order thinking skills. At random, the students were divided into two equal-sized groups. Members of the first group completed assignments using e-collaborative learning, working in subgroups, whereas members of the second group used e-individual learning, working independently. To examine the students' critical thinking and higher order thinking abilities, the researchers used the critical thinking scale and the higher order thinking skills test. The course was delivered via the Blackboard between January and May 2022. The critical thinking scale had three sections: define and adjust variables, hypothesis testing, and reasoning. Researchers examined the scale, which had a reliability coefficient of 0.91. The higher order thinking skills test consisted of 15 multiple-choice questions and 15 true/false questions, was graded by professionals, and had a reliability coefficient of 0.93.

The researchers used a statistical technique called "Multiple-group confirmatory factor analysis" (CFA) within a larger framework called "Structural Equation Model" (SEM) to determine whether using an online platform for collaborative learning had a meaningful effect on students' critical thinking skills. To see if the same online platform had any influence on students' higher order thinking skills, the researchers used a less difficult statistical technique known as a "T-test for independent samples."

Based on the study's findings, it can be said that using an online collaborative learning environment rather than an online individual learning environment has a more favorable impact on the development of higher order thinking skills and critical thinking. In particular, it was discovered that the E-collaborative learning environment in the first group had a positive influence on the three components of critical thinking - define and adjust variables, hypothesis testing, and reasoning - whereas the E-individual learning environment in the second group had less of an impact on these components. Additionally, the development of critical thinking and higher order thinking skills was also found to be positively impacted by all three parts of critical thinking, with the first group scoring significantly higher on a test of higher order thinking skills than the second group. It is significant to highlight that the study compared the effectiveness of the two learning settings using a multiple-group confirmatory factor analysis and a t-test for independent samples. The confirmatory factor analysis's fit statistics were strong, demonstrating the suitability of the critical thinking measurement model. The findings of the t-test revealed a substantial difference between the mean scores of the two groups on the test of higher order thinking skills, further demonstrating the value of the E-collaborative learning environment.

Table 1. Differences in higher order thinking skills among the two groups

Group	M	SD	M-Difference	T. Ratio	Sig.
1st group	26.17	2.96	4.63	5.26	.037
2nd group	21.54	4.18			

Overall, the results of this study show clear benefits of e-collaborative learning environments for the development of higher order thinking skills and critical thinking, indicating that e-

collaborative learning environments can have a positive effect on the development of higher order thinking skills and critical thinking and this research paper provides a good example to demonstrate how critical thinking ability of students is related to way of learning and teaching, and this way of learning and teaching can be enhanced via the proper integration of AI in education.

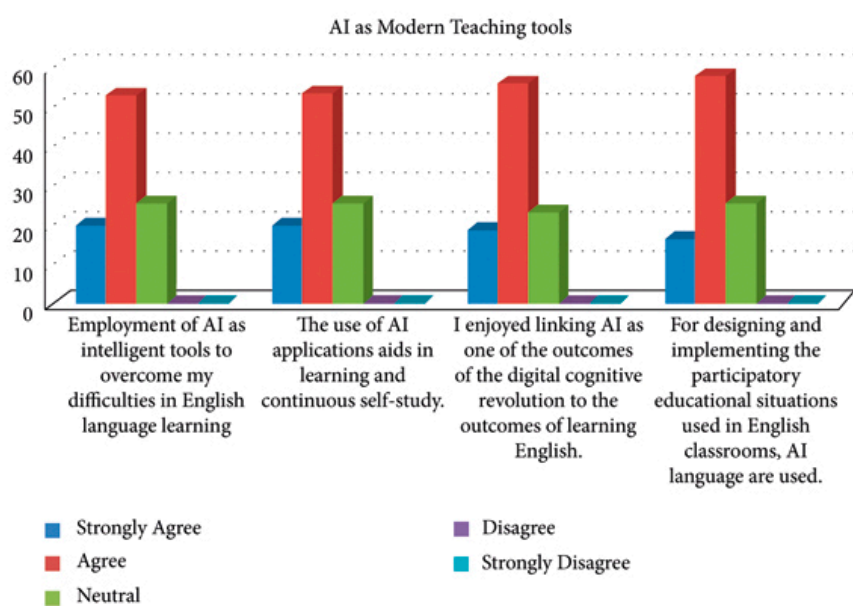
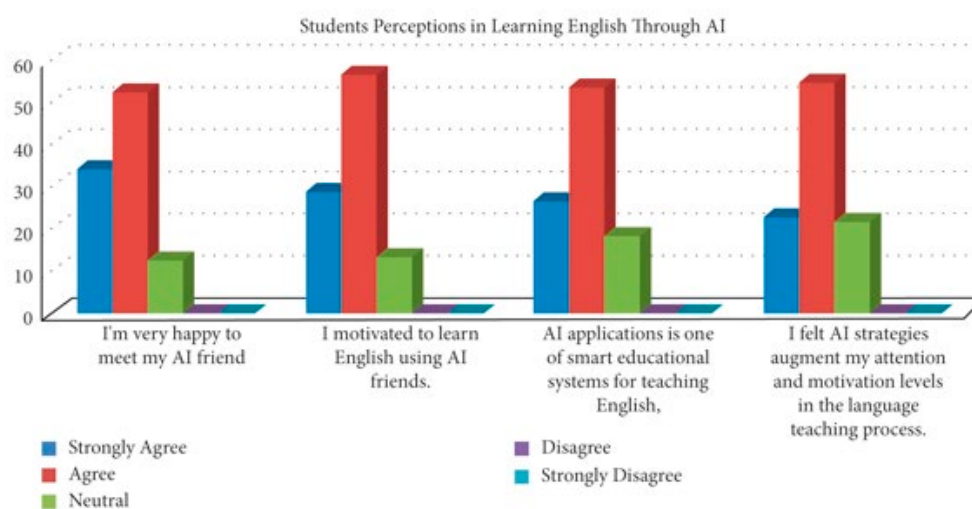
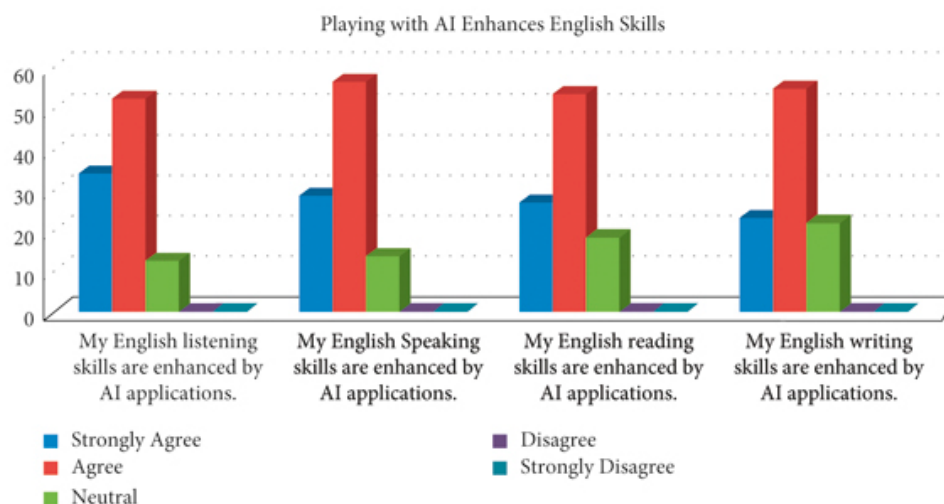
Lastly, the research paper (Muthmainnah et al., 2022), looked at how artificial intelligence (AI) can help people learning English as a second language strengthen their critical thinking skills. The researchers used a combination of online surveys and classroom observations to collect data from 453 individuals. Based on (Muthmainnah et al., 2022), the study found that using AI friend apps to teach people English improved their ability to think critically. Because to AI-based training, people's English language skills grew, as did their self-assurance, openness, and maturity. These are critical 21st-century skills that can be developed through AI-based training. This study used three different techniques to gather information about how critical thinking and English language learning are impacted by artificial intelligence (AI).

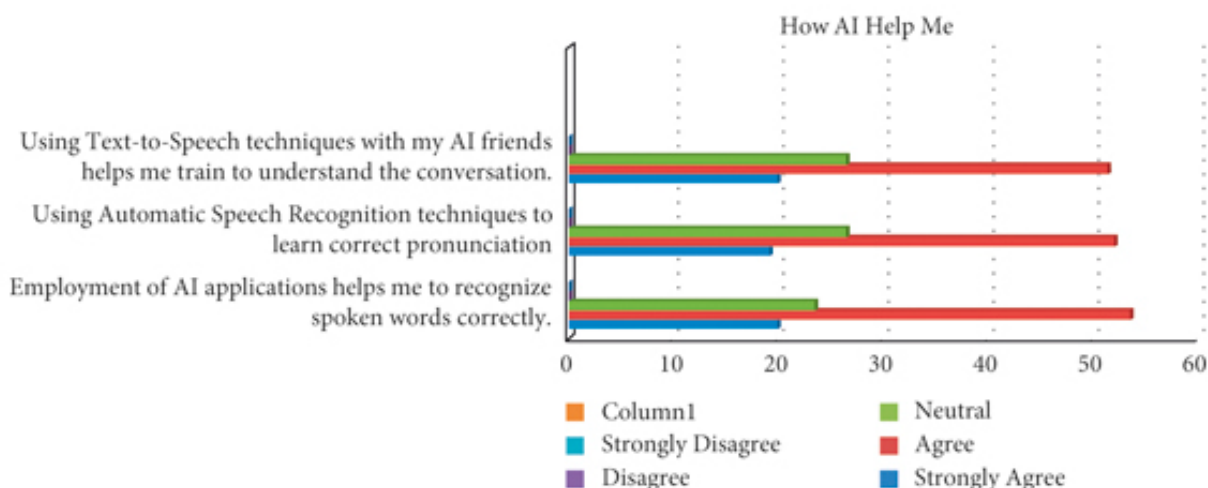
The first method involved giving college students a standardized language test. The test had 30 questions, of which 10 were essay questions and 20 were multiple-choice. The goal was to evaluate the students' knowledge of critical thinking and AI.

The second method used a survey that inquired about the students' opinions on AI. 24 questions on a Likert scale were included in the survey. The questionnaire asked how AI can help students become more proficient in English, pay attention, and be motivated, as well as how it can help them get past obstacles to learning English.

Classroom observation was the next technique used. The researcher observed the students' interactions with their AI friends for three months. The observations were carried out in order to validate the data acquired and to better understand the context of the data. In October, the three-month monitoring phase began. The data was collected using the OCIRAC phases, which stand for Observation, Collecting, Interacting, Reflecting, Associating, and Communicating. The procedure included observing English teaching materials, collecting data, interacting with AI companions to measure their impact on critical thinking skills, commenting on the experience, and communicating with other students. Students used a variety of strategies to practice their English throughout the teaching-learning process, including coming up with ideas, reviewing information, amassing evidence, writing draughts, talking with peers, and using it in authentic scenarios. To see if the use of AI has enhanced the participants' critical thinking abilities in English, they were given questionnaires that were identical to those they had previously filled out following the teaching-learning exercise.

The study's findings indicate that most students have favorable opinions about employing AI apps to learn English. Most students are motivated and content when utilizing AI friend tools, and they strongly agree or agree that AI applications are excellent educational systems for teaching English. The findings imply that students' critical thinking skills and English proficiency, including their listening, speaking, reading, and writing skills, are both enhanced by using AI. The study also found that students think artificial intelligence (AI) is a practical and contemporary technology for EFL and that it aids them in accurately recognizing spoken words. Through the use of text-to-speech and automatic voice recognition technologies with AI, students are trained to hear conversations and develop accurate pronunciation. Below are some of the charts of the findings:





Overall, the study found that AI can be useful teaching and learning tools for languages since they improve students' English language acquisition, and this study indicated that AI has a role to play in assisting people in developing their critical thinking skills.

Methodology and Methods

For this research study, mixed methodology approach is proposed to be used because the goal of the study is to get an in-depth understanding of the impact of AI on the development of critical thinking and problem-solving skills in university students. Mixed-methods research combines quantitative and qualitative research techniques to better comprehend a research subject. It is often used in interdisciplinary research projects or complex situations in the social, behavioral, and health sciences. By combining these two methodologies, researchers can learn more about their research topic in depth and gain greater insights (George, 2021).

For a thorough knowledge of research questions, mixed methodology integrates both quantitative and qualitative data. It allows for both statistical analysis and in-depth research of the issue, which can lead to a deeper understanding of complex topics.

In summary, this research study proposes the use of a mixed methods approach to gain a comprehensive understanding of the research topics and research questions. The study will involve the collection and analysis of both qualitative and quantitative data from multiple secondary data sources. Based on the qualitative and quantitative components of the available secondary data sources, the research will be carried out to investigate the research questions. To find themes and patterns in the qualitative data gathered from the three main secondary data sources, content analysis will be used. Statistical tools, such as straightforward percentage calculations and regression analysis, will be used to assess the quantitative data. The results from the various data sources will be combined and validated using data triangulation.

The software design implemented in this study is listed and described below:

1. The first study data source partially resolves question 1 and fully responds to questions 2 and 3. According to the first research data source (Muthmainnah et al., 2022), the qualitative components will involve investigation and comprehension of AI technology in language learning, particularly in the context of English language acquisition, use of literature reviews, evaluation and comparison of different scientific articles, collection of

data through questionnaires using Google Forms to gather qualitative data on the effectiveness of AI in language learning and identification of practical strategies for integrating of AI into language learning. On the other hand, the quantitative components will involve data collection through questionnaires and use of simple percentage calculation to determine the response of each respondent with the purpose of quantifying the responses of the participants and generate numerical data for statistical analysis in order to derive conclusions regarding effectiveness of AI in language learning and its impact on critical thinking.

To triangulate the data, the data collected from online survey and classroom observation will be integrated and validated. Moreover, researchers have pointed out on the requirement for review studies to better comprehend ethical and instructional methods.

2. The second data source mainly focuses on a quantitative study and responds to research question 2 by relating e-collaborating learning to integrated AI learning. According to the research (Alharbi et al., 2022), the quantitative components involve collection of data from critical thinking scale and higher-order thinking skills test. These data will be triangulated by statistical analysis of the outcome of the tests and compared between participating groups. Additionally, the researchers claimed that they adhered to all ethical guidelines when using human subjects in their research.
3. The third data source responds to the research question 1. According to the research (Zhu & Ren, 2022), the quantitative component will involve the use of questionnaire survey to collect data from number of teachers, students, school managers, family members of students from different education categories. And, qualitative components will involve by examining and observing roles cognition, human emotion, behaviour and their views on role identification.

To assess and come to conclusions on the impact of AI on role cognition in the educational system, the survey results will be triangulated using statistical analysis tools. Moreover, ethics factors, such as participant informed consent, confidentiality, and data protection, were considered throughout the research as per data source.

Ethical Considerations

It is crucial to take ethical principles into account when doing research. Therefore, throughout the research process, all ethical issues were considered, with all participants providing informed consent and maintaining their confidentiality, anonymity, and cultural sensitivity. To guarantee that the study was conducted in an ethical and professional manner, ethical principles were properly followed, and is also fully mentioned in all secondary data sources and research papers. In the development of research tools, ethics has played a critical role in ensuring that the methods utilised are ethical and respectful of participants.

Proposed Timeline (12 Weeks)

WEEK	ACTIVITY
Week 1	Choosing research Topic
Week 2	Selecting research questions, aim and objectives
Week 3	Annotated bibliography
Week 4	Gather information to write literature review
Week 5	Study methods of research
Week 6	Study methods of research
Week 7	Selecting right methodology
Week 8	Ethical considerations
Week 9	Data sampling
Week 10	Data integrations
Week 11	Writing proposal and presentation
Week 12	Submission

Conclusion

In conclusion, the goal of this study is to better understand how AI affects university students' capacity for critical thinking and problem-solving as well as how it can be integrated. With a focus on teaching and learning, it will especially examine the advantages of AI in education. Through this mixed methodology approach of the research, we hope to gain a deeper understanding of how AI is used in education and how it may be enhanced and incorporated to support students in learning these crucial skills while avoiding any potential drawbacks. Last but not least, the objective is to apply AI in a way that enhances student development and learning without compromising their ability to think critically and solve problems effectively.

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Appendices

Appendix-1

- A. My English listening skills are enhanced by AI applications.
1. Strongly Agree 2. Agree 3. Neutral 4. Disagree 5. Strongly Disagree
- B. My English-Speaking skills are enhanced by AI applications.
1. Strongly Agree 2. Agree 3. Neutral 4. Disagree 5. Strongly Disagree
- C. My English reading skills are enhanced by AI applications.
1. Strongly Agree 2. Agree 3. Neutral 4. Disagree 5. Strongly Disagree
- D. My English writing skills are enhanced by AI applications.
1. Strongly Agree 2. Agree 3. Neutral 4. Disagree 5. Strongly Disagree

Appendix-2

- A. Employment of AI as intelligent tools to overcome my difficulties in English language learning.
1. Strongly Agree 2. Agree 3. Neutral 4. Disagree 5. Strongly Disagree
- B. The use of AI applications aids in learning and continuous self-study.
1. Strongly Agree 2. Agree 3. Neutral 4. Disagree 5. Strongly Disagree
- C. I enjoyed linking AI as one of the outcomes of the digital cognitive revolution to the outcomes of learning English.
1. Strongly Agree 2. Agree 3. Neutral 4. Disagree 5. Strongly Disagree
- D. For designing and implementing the participatory educational situations used in English classrooms, AI language are used.
1. Strongly Agree 2. Agree 3. Neutral 4. Disagree 5. Strongly Disagree

Appendix-3

- A. My AI, as a modern tool, assists me in mastering the English language and gaining confidence in it.
1. Strongly Agree 2. Agree 3. Neutral 4. Disagree 5. Strongly Disagree
- B. My AI as a modern tool helps me solve problems of language learning, especially vocabulary.
1. Strongly Agree 2. Agree 3. Neutral 4. Disagree 5. Strongly Disagree
- C. I felt it was appropriate to practice my English with Replika, my AI friend.
1. Strongly Agree 2. Agree 3. Neutral 4. Disagree 5. Strongly Disagree
- D. I felt it was appropriate to practice my English with Mitsuku, my AI friend.
1. Strongly Agree 2. Agree 3. Neutral 4. Disagree 5. Strongly Disagree

Appendix-4

- A. Using Text-to-Speech techniques with my AI friends helps me train to understand the conversation.
 - 1. Strongly Agree 2. Agree 3. Neutral 4. Disagree 5. Strongly Disagree
- B. Using Automatic Speech Recognition techniques to learn correct pronunciation.
 - 1. Strongly Agree 2. Agree 3. Neutral 4. Disagree 5. Strongly Disagree
- C. Employment of AI applications helps me to recognize spoken words correctly.
 - 1. Strongly Agree 2. Agree 3. Neutral 4. Disagree 5. Strongly Disagree